Supplementary Materials

Transfer learning with simulated variants and calculated quantities for nanobody melting-temperature prediction

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Supplementary methods

Additional details for the diverse MD panel

The diverse panel was assembled from an archived SAbDab summary containing 2422 rows and the corresponding IMGT-renumbered structures [1]. For each PDB identifier, the workflow took the first heavy-chain entry whose chain identifier was present in the structure file. This deterministic rule gave 1177 initial structures. Of these, 1146 entered the 400-K manifest and 1143 yielded a usable sequence, topology, trajectory, and native-contact label.

The selected heavy chain was kept as an uncapped protein at pH 7.4, solvated in OPC water with 15 Å padding and 0.15 M NaCl, and described with ff19SB [2,3]. The 300-K branch comprised 1 ns NVT and 1 ns NPT with a 2-fs timestep. Two 1-ns restrained NVT stages at 400 K preceded 100 ns of unrestrained 400-K NVT production. Production used hydrogen-mass repartitioning to 4 amu, a 4-fs timestep, and output every 100 ps. OpenMM used PME electrostatics with a 1.0-nm cutoff, constraints on bonds to hydrogen, and a Langevin middle integrator with 1 ps^{-1} friction [4].

The diverse-panel contact calculation used backbone heavy-atom pairs separated by more than three residues, within 0.45 nm in the reference structure, and with at least one atom in Asp, Glu, Gln, Asn, Arg, Lys, or His. For trajectories with more than 300 frames, the label was averaged over the larger of the final 300 frames and the final one third of the trajectory; all frames were used for shorter trajectories. The processed table used 66–333 frames (median 333) and 28–728 contacts (median 173) per system. Sequence lengths were 58–461 residues (median 122). Supplementary Fig. S1 shows the relation between sequence length and the resulting raw Q value.

FEP provenance and charge-change check

The retained FEP tables contain Ala, Asp, Ile, and Gln scans. Their row counts were 116, 80, 119, and 120 for 1MEL and 105, 84, 113, and 107 for 4IDL, respectively. Row-level provenance was checked against the archived scan outputs; the available Phe scans were not included.

Of the 844 retained mutations, 284 change formal charge. For a cubic 70-Å box and solvent dielectric 78, we tested an analytical periodic net-charge correction of $0.086(\Delta q)^2$ kcal mol^{-1}. We applied the correction before repeating the sign reversal, robust scaling, Yeo–Johnson transform, and min–max scaling used for the model-facing labels. Corrected and uncorrected labels had Pearson $r = 0.99989$; the largest scaled-label shift was 0.008, and no row shifted by more than 0.02 (Supplementary Fig. S2b).

Model-selection audit

The completed search records contain 71 Stage-1 runs and 273 Stage-2 candidates: 115 with frozen ESM2 and 158 with fine-tuned ESM2. Each candidate was evaluated as a three-seed ensemble on the experimental Tm validation split; the selected setting was evaluated on the held-out test split with five seeds. The run seed controlled Python, NumPy, PyTorch, and CUDA randomness, as well as computed-row sampling and the computed-label train–monitoring split. Candidate scores, selected settings, final errors, label-count results, and figure-level provenance are retained as machine-readable tables in the analysis archive.

References

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[2] Tian, C., Kasavajhala, K., Belfon, K. A. A., Raguette, L., Huang, H., Migues, A. N., et al. ff19SB: amino-acid-specific protein backbone parameters trained against quantum mechanics energy surfaces in solution. Journal of Chemical Theory and Computation 16, 528–552 (2020). https://doi.org/10.1021/acs.jctc.9b00591

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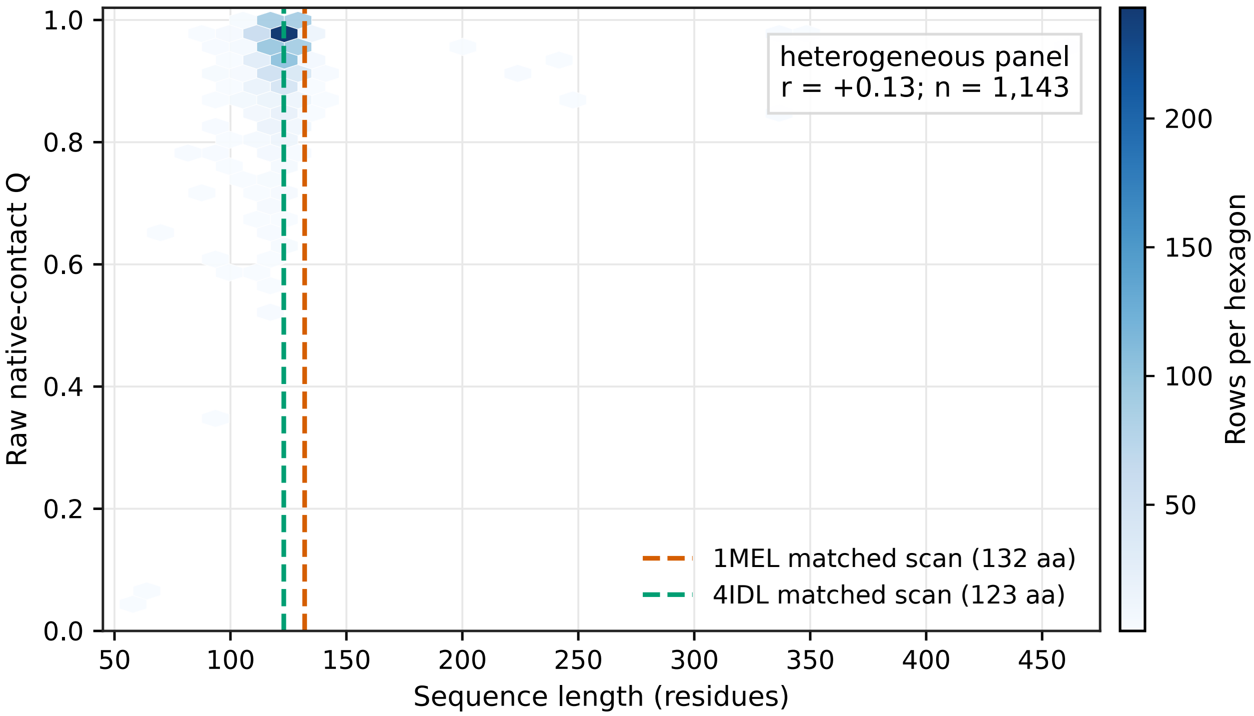

Figure S1 Sequence length and native-contact Q in the heterogeneous MD panel. Hexagons show the density of the 1143 processed rows. Dashed lines mark the fixed sequence lengths of the 1MEL and 4IDL matched mutation scans. In the heterogeneous panel, sequence length ranged from 58 to 461 residues and had a Pearson correlation of $r = +0.13$ with raw Q. This association identifies a possible shortcut but does not establish that length caused the transfer difference.

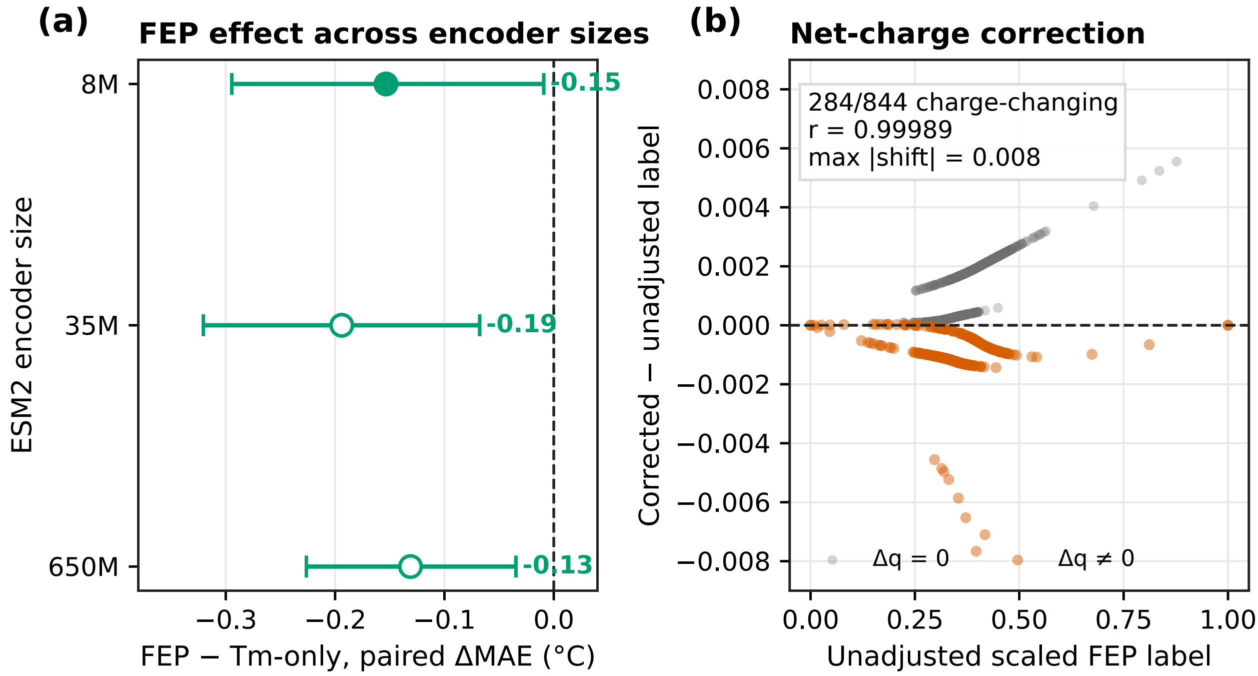

Figure S2 Robustness controls for the FEP result. **(a)** Paired FEP-minus-Tm-only MAE for fine-tuned ESM2 encoders. Negative values mean lower Tm error with FEP. The 8M point is the staged-selected five-seed model; the 35M and 650M points are exploratory fixed-configuration ten-seed controls and were not subjected to the same search. Whiskers are nominal 90% paired bootstrap intervals over the 396 test proteins. **(b)** Change in each scaled FEP label after the analytical periodic net-charge correction at dielectric 78. Of 844 labels, 284 were charge-changing. Corrected and uncorrected labels had Pearson $r = 0.99989$, and the maximum absolute shift was 0.008.